

Risksight:
Obstacle Awareness for FPV Drone Navigation
CS131
Zareef Shafquat

Original



Risk Overlay



Motivation & Problem

Frame 0



Frame 40



Frame 80



Frame 120



Frame 160



Problem:

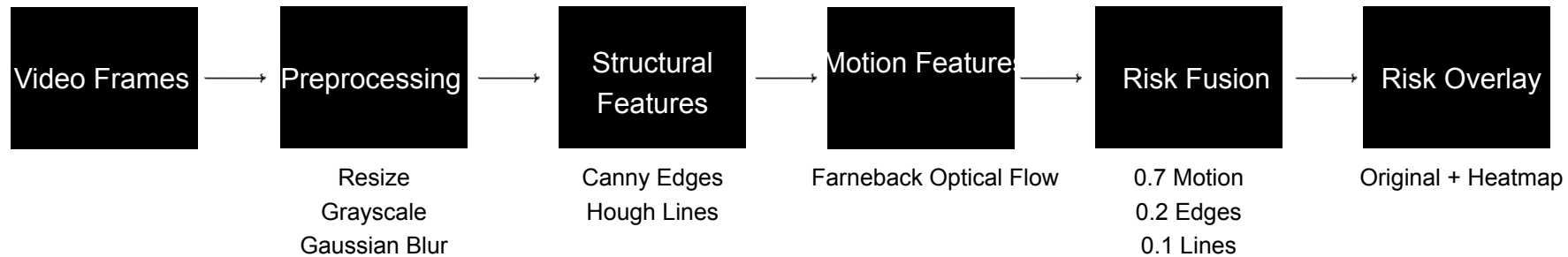
FPV drones operate at high speed

- Pilots only receive a monocular camera feed
- Judging obstacle proximity can be difficult
- Small mistakes can lead to collisions

Goal:

Highlight potentially risky regions directly from video using classical computer vision.

System Overview



System Overview Demo

Original Video Frame → Preprocessing → Structural Features

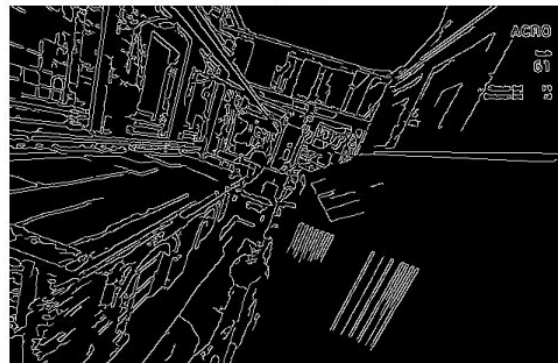
Original



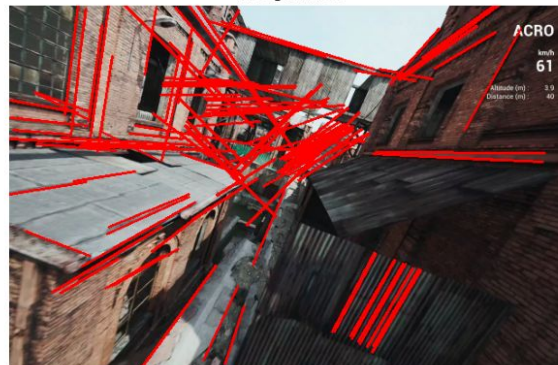
Grayscale + Blur



Canny Edges



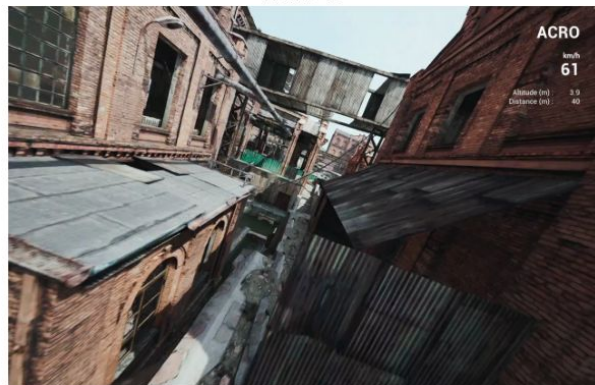
Hough Lines



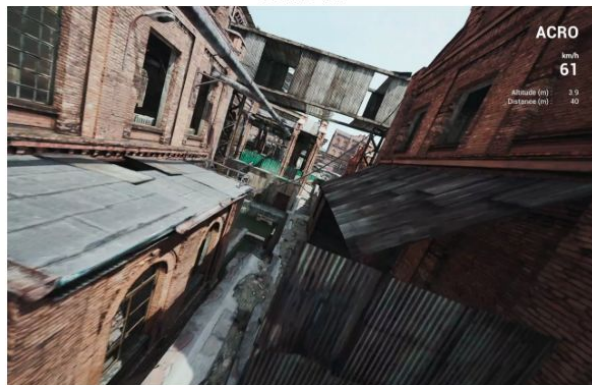
System Overview Demo (Continued)

Motion Estimation with Farneback Optical Flow

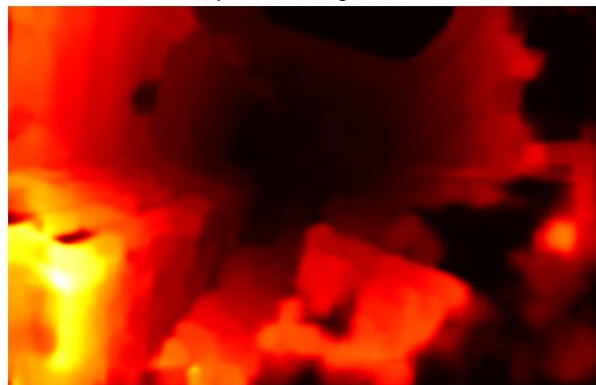
Frame 40



Frame 41



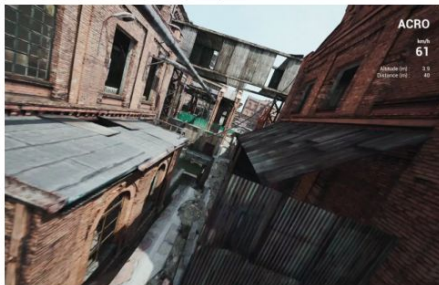
Optical Flow Magnitude



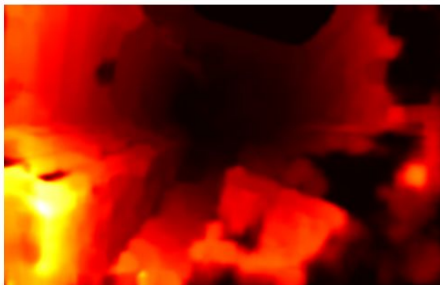
Dense Farneback optical flow estimates pixel motion between consecutive frames. Brighter regions correspond to larger motion magnitudes between consecutive frames.

Hybrid Risk Estimation

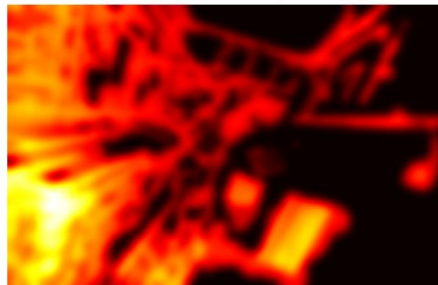
Original



Motion Score



Hybrid Risk Map



Hybrid Risk Overlay



$$\text{Risk} = 0.7(\text{Motion Score}) + 0.2(\text{Edge Map}) + 0.1(\text{Line Map})$$

- Motion acts as the primary cue because nearby objects generate stronger apparent motion.
- Edge features highlight object boundaries and structural detail.
- Hough line features emphasize large geometric structures such as walls, pipes, and building edges.

Results across Different Environments

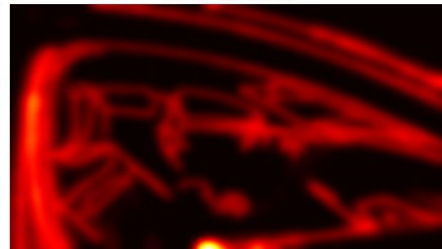
Original



Motion Score



Hybrid Risk Map



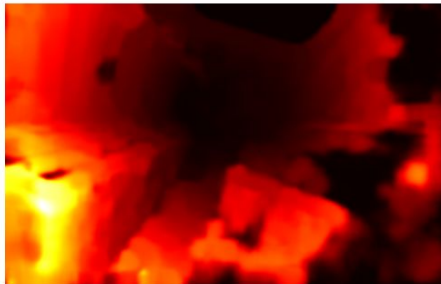
Hybrid Risk Overlay



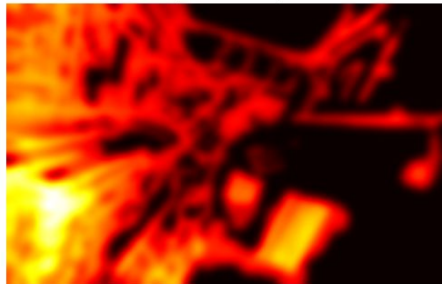
Original



Motion Score



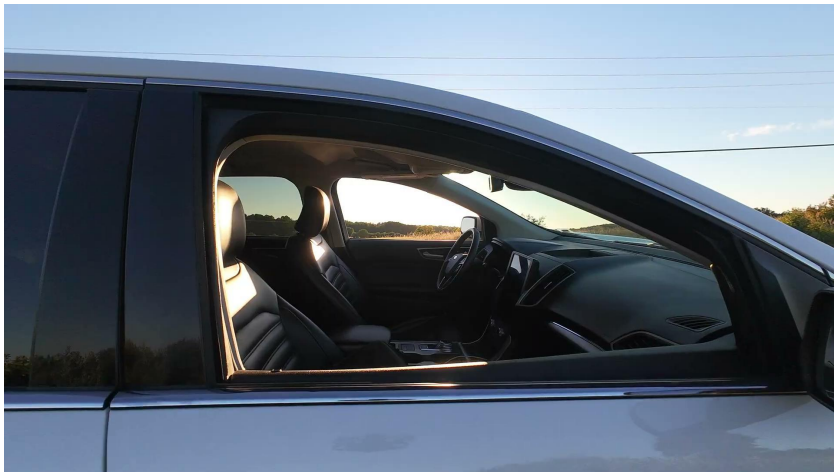
Hybrid Risk Map



Hybrid Risk Overlay



Demo Video - FPV Drone Through a Car



Demo Video - FPV Drone Through an abandoned factory

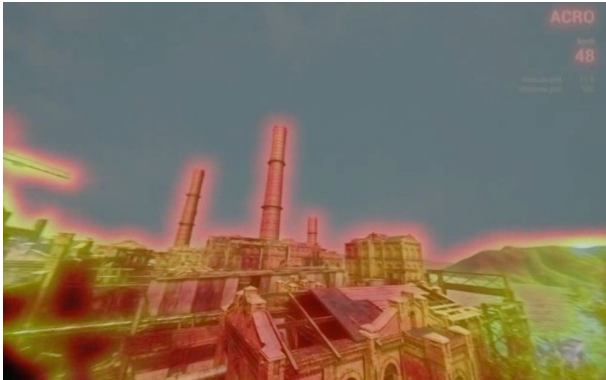


Limitations & Future Work



Limitations

- Optical flow may over-activate on textured terrain
- No explicit depth estimation
- Risk scores are based on image motion rather than semantic understanding
- Fusion weights are manually selected



Future Work

- Incorporate MiDaS or Depth Anything V2
- Explore adaptive weighting between motion and structure
- Improve robustness in open environments
- Evaluate on a larger and more diverse set of drone videos